

Multiple Choice (10 marks)

1. Which of the following is an incorrect statement about gel electrophoresis?
 - (a) DNA migrates from positive charge to negative charge.
 - (b) Smaller DNA travels faster.
 - (c) The DNA migrates only when the current is running.
 - (d) The longer the current is running, the farther the DNA will travel.
2. During the period when life is believed to have begun, the atmosphere on primitive Earth contained abundant amounts of all the following gases EXCEPT
 - (a) hydrogen
 - (b) methane
 - (c) nitrogen
 - (d) sulfur dioxide
 - (e) neon
3. Mutation of homeotic cluster genes often results in which of the following developmental defects in *Drosophila*?
 - (a) Complete absence of segmentation
 - (b) Hindrance in the development of the nervous system
 - (c) Inability to undergo metamorphosis
 - (d) Absence of a group of contiguous segments
 - (e) Tumor formation in imaginal discs
4. Which of the following would most likely provide examples of mitotic cell divisions?
 - (a) cross section of muscle tissue
 - (b) longitudinal section of a shoot tip
 - (c) longitudinal section of a leaf vein
 - (d) cross section of a fruit
 - (e) cross section of a leaf
5. Vinegar is classified as an acid. What does this tell you about the concentration of the hydrogen ions and the hydroxide ions?
 - (a) Vinegar has a lower concentration of hydrogen ions than hydroxide ions
 - (b) Vinegar has an excess of hydrogen ions with respect to hydroxide ions

- (c) Vinegar has a higher concentration of hydroxide ions than water
- (d) Vinegar has no hydroxide ions
- (e) Vinegar does not contain hydrogen ions

True / False (10 marks)

Answer True or False

- 6. True or False: Hemoglobin's affinity for O₂ increases as blood pH decreases, enhancing oxygen delivery to tissues.
- 7. True or False: The antibacterial properties of stomach acid prevent the stomach from being digested by its own secretions.
- 8. True or False: The shape of a wing is correlated with both the power and type of flight in birds.
- 9. True or False: Spindle fibers are composed of microtubules, which are essential for the proper segregation of chromosomes during mitosis.
- 10. True or False: Silk worm behavior is primarily controlled by external stimuli, such as light and temperature changes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the concept of stabilizing selection in relation to the average birth weight of human babies being approximately 3 kg. How does this phenomenon illustrate the principles of natural selection?
- 12. Discuss the concept of a flock of birds as an elementary form of society, focusing on their communication and cooperative behaviors in predator avoidance.

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